

# Insu Yun

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## Research Interests

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System security, software security, binary analysis, fuzzing, and applied cryptography.

## Education

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**Georgia Institute of Technology** Aug. 2015 – Dec. 2020  
Ph.D. in Computer Science  
Advisor: Dr. Taesoo Kim

**Korea Advanced Institute of Science and Technology (KAIST)** Sep. 2008 – Feb. 2015  
B.S. in Computer Science & Mathematics

## Work Experience

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**KAIST**, Daejeon, South Korea Feb. 2021 –  
Assistant Professor

**Microsoft Research**, Research Intern, Seattle, WA May. 2017 – Aug. 2017  
Contributed to REPT, a system that utilizes Intel Processor Trace to diagnose production failures  
Mentor: Weidong Cui

**Georgia Tech**, Research Assistant, Atlanta, GA Aug. 2015 – Dec. 2020

**Korean Cyber Command**, Software Developer, Seoul, Korea Apr. 2012 – Jan. 2014  
Served for the mandatory military service

## Publications

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### International Conferences (**Top-tier** and others)

1. **RTCon: Context-Adaptive Function-Level Fuzzing for RTOS Kernels (to appear)**  
Eunkyu Lee, Junyoung Park, and **Insu Yun**  
Proceedings of the 2026 Annual Network and Distributed System Security Symposium (**NDSS 2026**)  
February 2026
2. **CROSS-X: Generalized and Stable Cross-Cache Attack on the Linux Kernel**  
Dongok Kim\*, Juhyun Song\*, and **Insu Yun**  
Proceedings of the 32nd ACM Conference on Computer and Communications Security (**CCS 2025**)  
Taipei, Taiwan, October 2025
3. **Windows plays Jenga: Uncovering Design Weaknesses in Windows File System Security**  
Dong-uk Kim\*, JunYoung Park\*, Sanghak Oh, Hyoungshick Kim, and **Insu Yun**  
Proceedings of the 32nd ACM Conference on Computer and Communications Security (**CCS 2025**)  
Taipei, Taiwan, October 2025
4. **LLFuzz: An Over-the-Air Dynamic Testing Framework for Cellular Baseband Lower Layers**  
Tuan Dinh Hoang, Taekkyung Oh, CheolJun Park, **Insu Yun**, and Yongdae Kim

Proceedings of the 34th USENIX Security Symposium ([Security 2025](#))

Seattle, WA, August 2025

5. **Too Much of a Good Thing: (In-)Security of Mandatory Security Software for Financial Services in South Korea**

Taisic Yun, Suhwan Jeong, Yonghwa Lee, Seungjoo Kim, Hyoungshick Kim, **Insu Yun**, and Yongdae Kim

Proceedings of the 34th USENIX Security Symposium ([Security 2025](#))

Seattle, WA, August 2025

6. **FirmState: Bringing Cellular Protocol States to Shannon Baseband Emulation**

Suhwan Jeong, Beomseok Oh, Kwangmin Kim, **Insu Yun**, Yongdae Kim, and CheolJun Park

Proceedings of the 18th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2025)

Arlington, VA, June 2025

7. **Automated Attack Synthesis for Constant Product Market Makers**

Sujin Han, Jinseo Kim, Sung-Ju Lee, and **Insu Yun**

Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis 2025 ([ISSTA 2025](#))

Trondheim, Norway, June 2025

8. **Bridging the Gap between Real-World and Formal Binary Lifting through Filtered-Simulation**

Jihee Park, **Insu Yun**, and Sukyoung Ryu

Proceedings of the ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications 2025 ([OOPSLA 2025](#))

Singapore, October 2025

9. **RGFuzz: Rule-Guided Fuzzer for WebAssembly Runtimes**

Junyoung Park, Yunho Kim, and **Insu Yun**

Proceedings of the 46th IEEE Symposium on Security and Privacy ([Oakland 2025](#))

San Francisco, CA, May 2025

10. **BaseComp: A Comparative Analysis for Integrity Protection in Cellular Baseband Software**

Eunsoo Kim\*, Min Woo Baek\*, CheolJun Park, Dongkwan Kim, Yongdae Kim, and **Insu Yun**

Proceedings of the 32nd USENIX Security Symposium ([Security 2023](#))

Anaheim, CA, August 2023

11. **QueryX: Symbolic Query on Decompiled Code for Finding Bugs in COTS Binaries**

HyungSeok Han, JeongOh Kyea, Yonghwi Jin, Jinoh Kang, Brian Park, and **Insu Yun**

Proceedings of the 44th IEEE Symposium on Security and Privacy ([Oakland 2023](#))

San Francisco, CA, May 2023

12. **Fuzzing@Home: Distributed Fuzzing on Untrusted Heterogeneous Clients**

Daehee Jang, Ammar Askar, **Insu Yun**, Stephen Tong, Yiqin Cai, and Taesoo Kim

Proceedings of the 2022 International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2022)

October 2022

13. **DoLTEst: In-depth Downlink Negative Testing Framework for LTE Devices**

CheolJun Park\*, Sangwook Bae\*, BeomSeok Oh, Jiho Lee, Eunkyu Lee, **Insu Yun**, and Yongdae Kim

Proceedings of the 31th USENIX Security Symposium ([Security 2022](#))

Boston, MA, August 2022

(Acceptance rates: 18%, 256/1414)

14. **HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators**  
**Insu Yun**, Woosun Song, Seunggi Min, and Taesoo Kim

Proceedings of the 28th ACM Conference on Computer and Communications Security ([CCS 2021](#))

Seoul, South Korea, November 2021

(Acceptance rates: 22%, 196/880)

15. **Preventing Use-After-Free Attacks with Fast Forward Allocation**

Brian Wickman, Hong Hu, **Insu Yun**, Daehee Jang, JungWon Lim, Sanidhya Kashyap, and Taesoo Kim

Proceedings of the 30th USENIX Security Symposium ([Security 2021](#))

Vancouver, B.C., Canada, August 2021

(Acceptance rates: 19%, 246/1316)

16. **BaseSpec: Comparative Analysis of Baseband Software and Cellular Specifications for L3 Protocols**

Eunsoo Kim\*, Dongkwan Kim\*, Cheoljun Park, **Insu Yun**, and Yongdae Kim

Proceedings of the 2021 Annual Network and Distributed System Security Symposium ([NDSS 2021](#))

February 2021

(Acceptance rates: 15%, 87/578)

17. **Automatic Techniques to Systematically Discover New Heap Exploitation Primitives**

**Insu Yun**, Dhaval Kapil, and Taesoo Kim

Proceedings of the 29th USENIX Security Symposium ([Security 2020](#))

Boston, MA, August 2020

(Acceptance rates: 16%, 157/977)

18. **Fuzzing JavaScript Engines with Aspect-preserving Mutation**

Soyeon Park, Wen Xu, **Insu Yun**, Daehee Jang, and Taesoo Kim

Proceedings of the 41st IEEE Symposium on Security and Privacy ([Oakland 2020](#))

San Francisco, CA, May 2020

(Acceptance rates: 12%, 104/841)

[Nominated as a finalist in CSAW Best Applied Research Paper Award 2020](#)

19. **REPT: Reverse Debugging of Failures in Deployed Software**

Weidong Cui, Xinyang Ge, Baris Kasikci, Ben Niu, Upamanyu Sharma, Ruoyu Wang, and **Insu Yun** (alphabetical)

Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation ([OSDI 2018](#))

Carlsbad, CA, October 2018

(Acceptance rates: 18%, 47/257)

[Jay Lepreau Best Paper Award \(3 out of 257 submissions\)](#)

20. **QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing**

**Insu Yun**, Sangho Lee, Meng Xu, Yeongjin Jang, and Taesoo Kim

Proceedings of the 27th USENIX Security Symposium ([Security 2018](#))

Baltimore, MD, August 2018

(Acceptance rates: 19%, 100/524)

[Distinguished Paper Award \(5 out of 524 submissions\)](#)

21. **CAB-Fuzz: Practical Concolic Testing Techniques for COTS Operating Systems**

Su Yong Kim, Sangho Lee, **Insu Yun**, Wen Xu, Byoungyoung Lee, Youngtae Yun, and Taesoo Kim

Proceedings of the 2017 USENIX Annual Technical Conference ([ATC 2017](#))

Santa Clara, CA, July 2017

(Acceptance rates: 21%, 60/283)

22. **APISan: Sanitizing API Usages through Semantic Cross-checking**

**Insu Yun**, Changwoo Min, Xujie Si, Yeongjin Jang, Taesoo Kim, and Mayur Naik

Proceedings of the 25th USENIX Security Symposium ([Security 2016](#))

Austin, TX, August 2016

(Acceptance rates: 16%, 72/463)

[Nominated as a finalist in CSAW Best Applied Research Paper Award 2016](#)

## 23. **HDFI: Hardware-Assisted Data-Fow Isolation**

Chengyu Song, Hyungon Moon, Monjur Alam, **Insu Yun**, Byoungyoung Lee, Taesoo Kim, Wenke Lee, and Yunheung Paek

Proceedings of the 37th IEEE Symposium on Security and Privacy (**Oakland 2016**)

San Jose, CA, May 2016

(Acceptance rates: 13%, 55/413)

## 24. **Analyzing Security of Korean USIM-based PKI Certificate Service**

Shinjo Park, Suwan Park, **Insu Yun**, Dongkwan Kim, and Yongdae Kim

Proceedings of the 15th International Workshop on Information Security Applications (WISA 2014)

Jeju Island, Korea, August 2014

## 25. **Kargus: A Highly-scalable Software-based Intrusion Detection System**

Muhammad Jamshed, Jihyung Lee, Sangwoo Moon, **Insu Yun**, Deokjin Kim, Sungryoul Lee, Yung Yi, and KyoungSoo Park

Proceedings of the 19th ACM Conference on Computer and Communications Security (**CCS 2012**)

Raleigh, NC, October 2012

(Acceptance rates: 19%, 81/426)

## International Journal

## 26. **Scalable and Secure Virtualization of HSM with ScaleTrust**

Juhyeng Han, **Insu Yun**, Seongmin Kim, Taesoo Kim, Sooel Son, and Dongsu Han

IEEE/ACM Transactions on Networking (ToN)

November 2022

## Other Refereed Materials

## 27. **From the Vulnerability to the Victory: A Chrome Renderer 1-Day Exploit's Journey to v8CTF Glory**

Haein Lee, and **Insu Yun**

TyphoonCon 2024

Seoul, Korea, May 2024

## 28. **One shot, Triple kill: Pwning all three Google kernelCTF instances with a single 1-day Linux vulnerability**

Dongok Kim, Seunghyun Lee, and **Insu Yun**

POC 2023

Seoul, Korea, November 2023

## 29. **Compromising the macOS kernel through Safari by chaining six vulnerabilities**

Yonghwi Jin, Jungwon Lim, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2020)

Las Vegas, NV, August 2020

## 30. **AVPASS: Leaking and Bypassing Antivirus Detection Model Automatically**

Jinho Jung, Chanil Jeon, Max Wolotsky, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2017)

Las Vegas, NV, July 2017

## Thesis

## 31. **Concolic Execution Tailored for Hybrid Fuzzing**

**Insu Yun**

Ph.D. thesis, Georgia Institute of Technology

Atlanta, GA, December 2020

## Professional Activities

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### Technical Program Committee (International)

- Program Committee, *The International Conference on Cryptology and Network Security (CANS)*, 2025  
Program Committee, *Network and Distributed System Security Symposium (NDSS)*, 2025  
Program Committee, *Network and Distributed System Security Symposium (NDSS)*, 2024  
Program Committee, *IEEE Symposium on Security and Privacy (Oakland)*, 2024  
Program Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2023  
Program Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2022

### Journal Editor (International)

- Associate Editor, *ACM Transaction on Storage (ToS)*, 2024

### Others (International)

- Organization Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2024  
Artifact Evaluation Committee, *ACM Conference on Computer and Communications Security (CCS)*, 2023  
Artifact Evaluation Committee, *USENIX Security Symposium (Security)*, 2023  
Organization Committee, *ACM Conference on Computer and Communications Security (CCS)*, 2021

### Domestic Activities

- Security Task Force, *National Artificial Intelligence Strategy Committee*, 2025  
Advisory Boards, *HackTheon Sejong*, 2024–2025  
Organization Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

## Teaching Experience

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|                                                                             |             |
|-----------------------------------------------------------------------------|-------------|
| Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) | Fall 2024   |
| Information Security Laboratory (IS521 at KAIST)                            | Fall 2024   |
| Software Hacking Theory and Practice (EE517 at KAIST)                       | Spring 2024 |
| • Evaluation – Average: 4.92 / 5                                            |             |
| Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) | Fall 2023   |
| • Evaluation – Average: 4.57 / 5                                            |             |
| Software Hacking Theory and Practice (EE517 at KAIST)                       | Spring 2023 |
| • Evaluation – Average: 4.54 / 5                                            |             |
| Programming Structures for Electronical Engineering (EE209 at KAIST)        | Fall 2022   |
| • Evaluation – Average: 4.65 / 5                                            |             |
| Software development environment and tools practice (EE485-A at KAIST)      | Fall 2022   |
| • Evaluation – Average: 4.43 / 5                                            |             |
| Software Security (EE595-B at KAIST)                                        | Spring 2022 |
| • Evaluation – Average: 5 / 5                                               |             |
| Programming Structures for Electronical Engineering (EE209 at KAIST)        | Fall 2021   |
| • Evaluation – Average: 4.34 / 5                                            |             |
| Software development environment and tools practice (EE485-A at KAIST)      | Fall 2021   |
| • Evaluation – Average: 4.34 / 5                                            |             |
| Software Security (EE595-B at KAIST)                                        | Spring 2021 |
| • Evaluation – Average: 4.9 / 5                                             |             |

## Honors & Awards

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### Academic awards

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|--------------------------------------------------------------------------------------|-----------------|
| Song-Am Future Researcher Award                                                      | KAIST Feb. 2025 |
| Technology Innovation Awards, KAIST College of Engineering                           | Dec. 2024       |
| Best Teaching Award, KAIST Electrical Engineering                                    | Oct. 2024       |
| Prize for Excellence in Teaching, KAIST                                              | Feb. 2024       |
| Frontiers of Science Award, The First International Congress of Basic Science (ICBS) | July. 2023      |
| Best Teaching Award, KAIST Electrical Engineering                                    | Sep. 2021       |
| Jay Lepreau Best Paper Award, USENIX OSDI 2018                                       | Aug. 2018       |
| Distinguished Paper Award, USENIX Security 2018                                      | Aug. 2018       |

## Hacking competitions

|                                                |           |
|------------------------------------------------|-----------|
| DEFCON 26 CTF, 1st place (Team DEFKOR00T)      | Aug. 2018 |
| DEFCON 24 CTF, 3rd place (Team DEFKOR)         | Aug. 2016 |
| DEFCON 23 CTF, 1st place (Team DEFKOR)         | Aug. 2015 |
| Whitehat contest 2014, 1st place (Team SysSec) | Nov. 2014 |
| DEFCON 22 CTF, 10th place (Team GoN)           | Aug. 2014 |
| SECCON CTF 2014, 1st place (TOEFL Beginner)    | Feb. 2014 |
| Codegate CTF 2012, 3rd place (Team GoN)        | Apr. 2012 |
| Secuinside CTF, 3rd place (Team GoN)           | Oct. 2011 |
| ISEC CTF, 1st place (Team GoN)                 | Sep. 2011 |
| DEFCON 18 CTF, 3rd place (Team GoN)            | Aug. 2010 |
| Codegate CTF 2010, 5th place (Team GoN)        | Apr. 2010 |
| KISA HDCON, Gold Medal, 2nd place (Team GoN)   | May 2009  |
| Codegate CTF 2009, 4th place (Team GoN)        | Apr. 2009 |

## Scholarships

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|---------------------------------------------------------------------|-----------------------|
| National Research Foundation of Korea Scholarship for Undergraduate | Mar. 2008 – Dec. 2013 |
|---------------------------------------------------------------------|-----------------------|

## Others

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| DARPA AIXCC, 1st place (Team Atlanta), \$4M award                            | Aug. 2025 |
| Cyber Security Challenge, 2nd place (Team HackingLab), \$400K research grant | 2023      |
| DARPA Cyber Grand Challenge, Finalist (Team Disekt)                          | Aug. 2016 |
| KISA Bug Bounty Program's Hall of Fame                                       | 2013      |

## Vulnerability Discovery Reward (aka Bug bounty)

To summarize, \$244.6K (by my students) and \$92.8K (by myself) bug bounties are awarded so far.

### By my students

|                                                                                                     |           |
|-----------------------------------------------------------------------------------------------------|-----------|
| Pwn2Own - Microsoft Edge and Google Chrome (\$145K), ZDI, by SeunHyun Lee                           | Mar. 2024 |
| v8CTF - CVE-2023-6702 (\$10K), Google, by Haein Lee                                                 | Jan. 2024 |
| kernelCTF - CVE-2023-3390 (\$67.8K), Google, by Dongok Kim and SeunHyun Lee                         | Oct. 2023 |
| Type confusion in V8 (\$7K), Google, by Haein Lee                                                   | Mar. 2023 |
| NAS authentication bypass in Exynos (\$14.8K), Samsung Electronics, by Eunsoo Kim and CheolJun Park | Feb. 2022 |

### By myself

|                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------|-----------|
| PSV-2021-0304: afpd auth bypass (\$300), NETGEAR                                                       | Mar. 2021 |
| Pwn2Own Apple Safari with a kernel privilege escalation (\$70K), ZDI, with Yonghwi Jin and Jungwon Lim | Mar. 2020 |
| Apple Safari sandbox escape (\$20K), Apple                                                             | Dec. 2019 |
| Three integer overflow vulnerabilities in PHP (\$1.5K), the Internet Bug Bounty                        | Jun. 2016 |

## Patents

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### International

#### 2. Security analysis system and method based on negative testing for protocol implementation of LTE device (Pending)

Inventors: Yongdae Kim, Cheoljun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Application date: 2022.10.05

Application number: 17960246

Country: US

#### 1. Reverse debugging of software failures

Inventors: Weidong Cui, Xinyang Ge, Baris Kasikci, Cengiz Can, Ben Niu, Ruoyu Wang, Insu Yun

Registration date: 10565511

Patent number: 2020.02.18

Country: US

### Domestic

#### 3. Security analysis system and method based on negative testing for protocol implementation of LTE device

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, BeomSeok Oh, Jiho Lee, Eunkyu Lee, Insu Yun

Registration date: 10-2514797-0000

Patent number: 2023.03.23

Country: Korea

#### 2. Method and system for automatically analyzing bugs in cellular baseband software using comparative analysis based on cellular specifications

Inventors: Yongdae Kim, Eunsoo Kim, Dongkwan Kim, CheolJun Park, Insu Yun

Registration date: 10-2546946-0000

Patent number: 2023.06.20

Country: Korea

#### 1. Methods and systems for key management service provision (Pending)

Inventors: Dongsoo Han, JuHyeng Han, Insu Yun

Application date: 10-2021-0154174

Application number: 2021.11.10

Country: Korea

## Invited Talks

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### International

Title: How to build Skynet — a system that hacks systems

**Keynote** speech at TyphoonCon, Seoul, Korea

Jun. 2023

Title: HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators

Presented at ACM CCS 2021, Online

Nov. 2021

Title: Automatic Techniques to Systematically Discover New Heap Exploitation Primitives

Presented at USENIX Security 2020, Online

Aug. 2020

|                                                                                                                                    |           |
|------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Title: QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing<br>Presented at USENIX Security 2018, Baltimore, MD | Aug. 2018 |
| Title: APISan: Sanitizing API Usages through Semantic Cross-checking<br>Presented at USENIX Security 2016, Austin, TX              | Aug. 2016 |

## Domestic

|                                                                                                                                                                                                                                                                                                                                       |                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Title: The Dark and Bright Sides of AI in Cybersecurity<br>Seminar at KAIST-Cyber Security Policy Program<br>Seminar at Samsung SDS, Seoul, Korea<br>Seminar at National Intelligence Service, Seoul, Korea                                                                                                                           | Sep. 2025<br>Sep. 2025<br>Sep. 2025                           |
| Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System<br>Seminar at Cyber Summit Korea 2025, Seoul, Korea<br>Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea<br>Seminar at ENKI WhiteHat, Seoul, Korea<br>Seminar at LIGNex1, Seoul, Korea<br>Seminar at KCSCON, Seoul, Korea | Sep. 2025<br>Aug. 2025<br>Sep. 2025<br>Sep. 2025<br>Sep. 2025 |
| Title: AIxCC: The Future of Cybersecurity with LLMs<br>Seminar at ETRI, Daejeon, Korea<br>Seminar at Moon Sool Graduate School of Future Strategy, Daejeon, Korea<br>Seminar at Future Security Tech Forum, Jeju, Korea<br>Seminar at National Security Research Institute (NSRI), Daejeon, Korea                                     | Nov. 2024<br>Nov. 2024<br>Nov. 2024<br>Sep. 2024              |
| Title: 2024 Security Strategy: Polarization<br>Seminar at Defense Counterintelligence Command, Gwacheon, Korea                                                                                                                                                                                                                        | May. 2024                                                     |
| Title: Building Automated Hacking Systems<br>Seminar at POSTECH, Pohang, Korea                                                                                                                                                                                                                                                        | Nov. 2023                                                     |
| Title: Trends in Security Vulnerabilities of Low Earth Orbit Satellites<br>Presented at ETRI, Daejeon, Korea                                                                                                                                                                                                                          | Aug. 2023                                                     |
| Title: Academic Research from Offensive Research<br>Presented at Samsung, Seoul, Korea                                                                                                                                                                                                                                                | Aug. 2023                                                     |
| Title: Human-friendly binary analysis<br>Presented at ETRI, Daejeon, Korea<br>Presented at Korea Computer Congress (KCC), Seoul, Korea                                                                                                                                                                                                | Nov. 2023<br>Jun. 2023                                        |
| Title: Exploit in the wild<br>Presented at ETRI, Daejeon                                                                                                                                                                                                                                                                              | Jun. 2023                                                     |
| Title: Hacking 101<br>Presented at WISC, Seoul                                                                                                                                                                                                                                                                                        | Sep. 2022                                                     |
| Title: Attack and Defenses for Heap Vulnerabilities in 2022<br>Seminar at ETRI, Daejeon                                                                                                                                                                                                                                               | Apr. 2022                                                     |
| Title: Comparative Analysis of Baseband Software and Cellular Specifications for Finding Vulnerabilities<br>Seminar at UNIST, Ulsan<br>Seminar at Security@KAIST, Online<br>Seminar at Cyber Operations Command, Seoul                                                                                                                | May. 2023<br>Jun. 2022<br>Jun. 2022                           |
| Title: Scalable and Automatic Vulnerability Discovery Beyond Random Testing<br>Seminar at Seoul National University, Seoul, Korea, Mar. 2019                                                                                                                                                                                          |                                                               |
| Title: Memory Allocator Security<br>Presented at Best of Best (BoB), Seoul<br>Presented at Computer System Society Conference (CSC), Pyeongchang                                                                                                                                                                                      | Feb. 2023<br>Feb. 2023                                        |



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|-----------------------------------------------------------------|-----------|
| Seminar at UNIST, Online                                        | May. 2022 |
| Seminar at Yonsei university, Online                            | Apr. 2022 |
| Seminar at Sungkyunkwan university, Online                      | Apr. 2022 |
| Seminar at ETRI, Daejeon                                        | Jan. 2022 |
| Seminar at National Security Research Institute (NSRI), Daejeon | Dec. 2021 |
| Seminar at Securty@KAIST, Online                                | Nov. 2021 |
| Seminar at KAIST GSIS, Online                                   | Nov. 2021 |
| Title: Browser Security: Hacking & Research                     |           |
| Presented at Open Theori Research Seminar #6, Online            | Dec. 2021 |
| Seminar at Hanyang University, Online                           | Nov. 2021 |
| Presented at KR Becks Meetup #1 by LINE, Online                 | Aug. 2021 |
| Seminar at Security@KAIST, Online                               | Jun. 2021 |

## Grants

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To summarize, \$2 million is awarded, and my portion is \$1.42 million. Please note that I have accounted for the exchange rate of 1,000 won to one dollar.

|                                                                                                                           |               |
|---------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Research on building an open source kernel security model</b>                                                          | 24.04 – 24.10 |
| Agency/Company: NRF                                                                                                       |               |
| Money: \$54.5K                                                                                                            |               |
| Role: PI                                                                                                                  |               |
| <b>Research on cybersecurity technologies for the future</b>                                                              | 24.03 – 24.12 |
| Agency/Company: Future science academy                                                                                    |               |
| Money: \$140K                                                                                                             |               |
| Role: PI                                                                                                                  |               |
| <b>Revisiting IoT threat models for smart cities and developing a vulnerability analysis system based on these models</b> | 24.01 – 25.12 |
| Agency/Company: IITP                                                                                                      |               |
| Money: \$400k                                                                                                             |               |
| Role: PI                                                                                                                  |               |
| <b>Building a system to assist variant analysis for browsers</b>                                                          | 23.06 – 24.05 |
| Agency/Company: NRF                                                                                                       |               |
| Money: \$65K                                                                                                              |               |
| Role: PI                                                                                                                  |               |
| <b>Generating a security model based on JavaScript intermediate language</b>                                              | 23.04 – 23.10 |
| Agency/Company: NSRI                                                                                                      |               |
| Money: \$54.5K                                                                                                            |               |
| Role: PI                                                                                                                  |               |
| <b>Verifying security threats in open-source operating systems</b>                                                        | 23.04 – 23.10 |
| Agency/Company: NSRI                                                                                                      |               |
| Money: \$54.5K                                                                                                            |               |
| Role: PI                                                                                                                  |               |
| <b>An automated framework that generates exploit for multi-type kernel bugs</b>                                           | 23.02 – 23.11 |
| Agency/Company: CISC                                                                                                      |               |
| Money: \$100K                                                                                                             |               |
| Role: PI                                                                                                                  |               |
| <b>Browser fuzzing with formal verification for cross architectures</b>                                                   | 22.09 – 23.09 |
| Agency/Company: NRF                                                                                                       |               |
| Money: \$110K                                                                                                             |               |
| Role: PI                                                                                                                  |               |

|                                                                                                                                         |               |
|-----------------------------------------------------------------------------------------------------------------------------------------|---------------|
| <b>Building test suites for validating vulnerability detection</b>                                                                      | 22.08 – 22.11 |
| Agency/Company: ETRI                                                                                                                    |               |
| Money: \$27.3K                                                                                                                          |               |
| Role: PI                                                                                                                                |               |
| <b>Generating a security model based on JavaScript security analysis</b>                                                                | 22.04 – 22.10 |
| Agency/Company: NSRI                                                                                                                    |               |
| Money: \$54.5K                                                                                                                          |               |
| Role: PI                                                                                                                                |               |
| <b>Developing techniques for collection and integrated analysis of automotive systems through event-based experimental systems</b>      | 22.04 – 23.12 |
| Agency/Company: Dankook university                                                                                                      |               |
| Money: \$300K $\times$ 0.5                                                                                                              |               |
| Role: PI working with Prof. Yujip Won                                                                                                   |               |
| <b>6G security</b>                                                                                                                      | 21.08 – 23.09 |
| Agency/Company: Samsung Electronics                                                                                                     |               |
| Money: \$200K $\times$ 0.2                                                                                                              |               |
| Role: Co-PI with Prof. Yongdae Kim                                                                                                      |               |
| <b>DRAM security</b>                                                                                                                    | 21.07 – 24.06 |
| Agency/Company: Samsung Electronics                                                                                                     |               |
| Money: \$180K $\times$ 0.2                                                                                                              |               |
| Role: Co-PI with Prof. Yongdae Kim                                                                                                      |               |
| <b>Systematic and precise transformation of the Qualcomm Hexagon architecture into intermediate representations for binary analysis</b> | 21.06 – 22.05 |
| Agency/Company: NRF                                                                                                                     |               |
| Money: \$46.7K                                                                                                                          |               |
| Role: PI                                                                                                                                |               |
| <b>Automatically generating a security model for discovering web browser vulnerabilities</b>                                            | 21.04 – 21.10 |
| Agency/Company: NSRI                                                                                                                    |               |
| Money: \$54.5K                                                                                                                          |               |
| Role: PI                                                                                                                                |               |
| <b>Developing a scalable cyber reasoning system (Start-up)</b>                                                                          | 21.02 – 24.12 |
| Agency/Company: KAIST                                                                                                                   |               |
| Money: \$150K                                                                                                                           |               |
| Role: PI                                                                                                                                |               |

## Advising and Mentoring

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### Ph.D. Students

|                 |             |
|-----------------|-------------|
| - Eunkyu Lee    | Fall 2023   |
| - Minwoo Baek   | Spring 2024 |
| - Junyeong Park | Spring 2024 |

### Ph.D./M.S Students

|             |             |
|-------------|-------------|
| - Haein Lee | Spring 2022 |
|-------------|-------------|

### M.S. Students

|                 |             |
|-----------------|-------------|
| - Dongok Kim    | Spring 2023 |
| - Seunggi Min   | Fall 2023   |
| - Donguk Kim    | Spring 2024 |
| - Donghyeon Kim | Spring 2024 |
| - Wonyoung Kim  | Spring 2024 |

- Hyeon Heo

Spring 2024

## **Alumni**

- Wonyeong Jung, 78ResearchLab

M.S. in Fall 2024

- Hyunsik Jeong (Co-advising with Yongdae Kim), S2W

M.S. in Fall 2021

- Hyungseok Han (Co-advising with Yongdae Kim), Postdoc at Georgia Tech

Ph.D. in Fall 2022